

Satellite Imagery–Based Property Valuation

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1. Overview

The objective of this project is to develop a robust property valuation system that combines traditional tabular housing data with satellite imagery to predict residential property prices. While conventional models rely heavily on structured features such as house size, location, and construction quality, they often fail to capture environmental and neighborhood context.

To overcome this limitation, a multimodal regression pipeline was designed that integrates satellite imagery with structured data. Satellite images provide visual cues such as greenery, urban density, road connectivity, and surrounding infrastructure. The project also emphasizes model explainability using Grad-CAM to understand how visual features influence predictions.

2. Dataset Description

2.1 Tabular Data

The tabular dataset contains historical housing records with the following key features:

- Number of bedrooms and bathrooms
- Living area (`sqft_living`) and lot size (`sqft_lot`)
- Construction quality (`grade`) and condition
- Geographic coordinates (`latitude`, `longitude`)
- Neighborhood statistics (`sqft_living15`, `sqft_lot15`)

The target variable is **property price**, which is log-transformed during training to stabilize variance.

2.2 Satellite Imagery Data

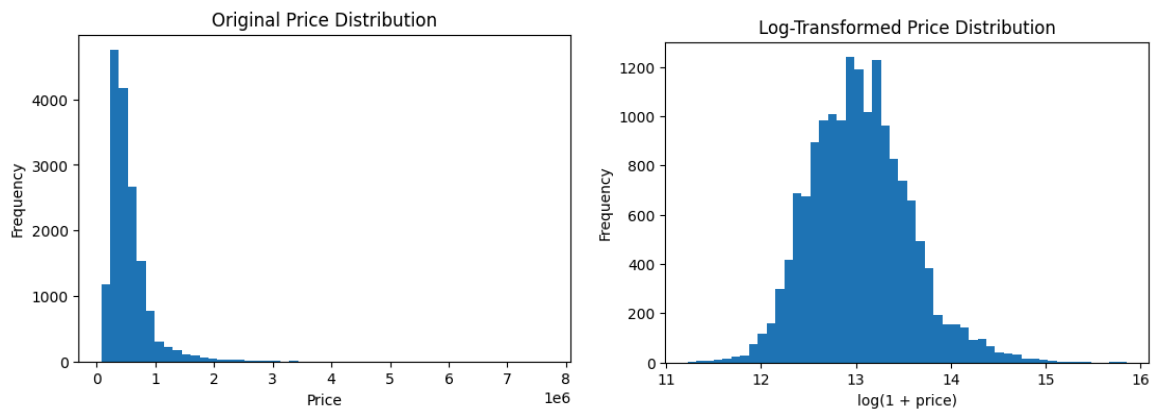
Satellite images are programmatically fetched using the Mapbox Static Images API. Each image is centered at the latitude–longitude coordinates of the corresponding property and captures environmental context such as green cover, nearby roads, and urban layout.

3. Exploratory Data Analysis (EDA)

Exploratory analysis was performed to understand the distribution of property prices and relationships between features.

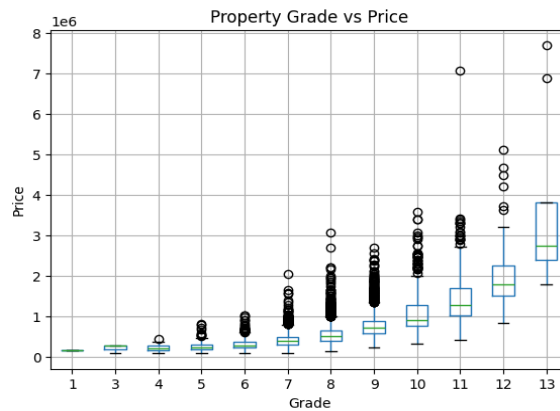
3.1 Price Distribution

- Property prices are right-skewed with a small number of high-value properties.
- A logarithmic transformation ($\log(1+p)$) was applied to improve model stability.



3.2 Feature Relationships

- Living area and construction grade show strong positive correlation with price.
- Neighborhood features influence valuation by capturing local density effects.



Larger living areas generally correspond to higher property prices, though price variability increases significantly for very large homes, indicating the influence of additional qualitative factors beyond size and property price increases steadily with higher construction grade, showing a strong positive relationship between overall build quality and market value.

3.3 Satellite Image Samples

Sample satellite images were visualized to observe variation in environmental context across properties.





4. Modeling Approach

4.1 Tabular Baseline Model

A Random Forest Regressor was trained using only structured tabular features. This model serves as a strong baseline and provides reliable predictions when satellite imagery is unavailable.

Validation Performance:

- RMSE $\approx$ 145,000
- $R^2 \approx 0.83$

4.2 Multimodal Model

The multimodal architecture consists of:

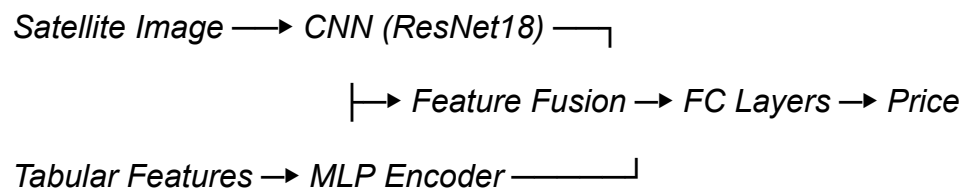
- **Image Encoder:** ResNet18 (ImageNet pretrained, frozen)
- **Tabular Encoder:** Multi-layer perceptron (MLP)
- **Fusion Strategy:** Late fusion via feature concatenation
- **Loss Function:** Mean Squared Error (log-price space)

This model jointly learns from visual and numeric inputs.

5. Architecture Diagram

The overall pipeline consists of:

1. Tabular feature encoder
2. CNN-based image encoder
3. Feature fusion layer
4. Regression head for final price prediction

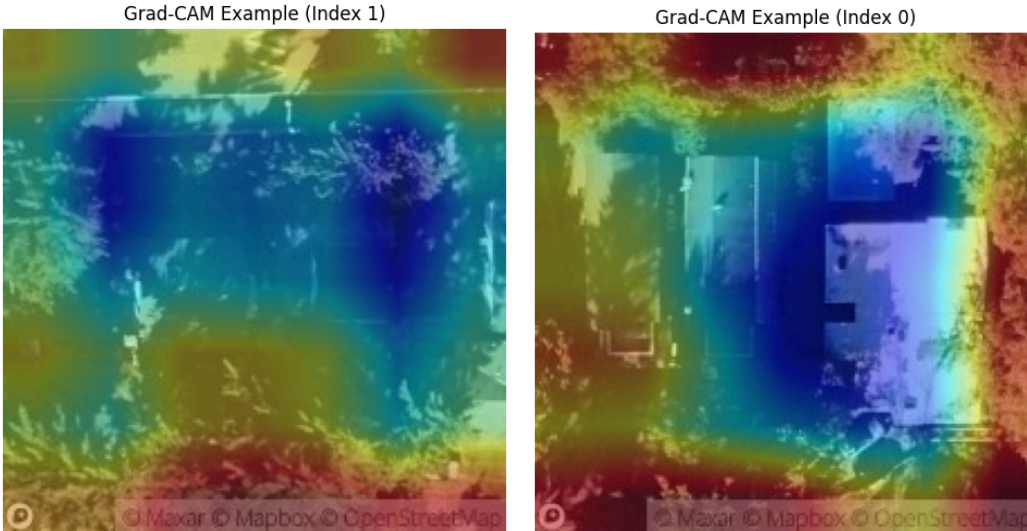


6. Explainability Using Grad-CAM

Grad-CAM was applied to the final convolutional layer of the image encoder to visualize which regions of satellite imagery influenced model predictions.

Key observations:

- Green cover and vegetation often contribute positively to price predictions.
- Urban density patterns and road networks are highlighted in higher-value regions.
- Infrastructure and surrounding layout play a role in valuation.



7. Inference Strategy and Results

Due to limited availability of satellite images, a **hybrid inference strategy** was implemented:

- When satellite imagery is available and reliable, the multimodal model is used.
- When imagery is missing or unreliable, predictions fall back to the tabular-only model.

This strategy ensures stable predictions for all test samples while demonstrating the feasibility of multimodal learning.

8. Conclusion

This project demonstrates how satellite imagery can be integrated into traditional property valuation models to incorporate environmental context and improve interpretability. While structured data remains the dominant signal, visual features provide complementary insights. The hybrid design reflects real-world deployment constraints and balances performance, robustness, and explainability.